

Trustworthiness Engineering in Telecommunications

Collection Processes for

Sensor Ground Truth and Continuous Value Telemetry

Jeffrey Walk - February 2026

Overview

Underlying Assertion: Large-scale telecommunications networks are transitioning from probabilistic AI systems to Trustworthiness Engineering frameworks that anchor digital operations in physical and logical reality. This document emphasizes the critical collection processes that enable this transition, focusing on how organizations capture deterministic sensor ground truth and implement continuous value telemetry to govern AI agents effectively.

The foundation of trustworthy AI in telecommunications rests on two parallel data collection architectures: (1) **sensor networks** that establish what is physically and logically true about network state, and (2) telemetry systems that continuously measure drift between human intent and AI execution. Without robust collection processes, AI agents operate on stale or incomplete information, creating systemic risk of confident but incorrect decisions.

1. The Collection Architecture: Overview

The Two-Pipeline Model

Trustworthiness Engineering requires two distinct but complementary data collection pipelines:

- **Ground Truth Collection Pipeline:** Captures deterministic sensor data about the physical and logical state of network infrastructure. This includes optical power measurements, protocol state machines, flow statistics, and environmental conditions. These sensors answer the question: "What is actually true right now?"
- **Value Telemetry Collection Pipeline:** Captures continuous measurements of service quality, SLA compliance, and business metrics that represent stakeholder value. These telemetry streams answer the question: "Is the system delivering what humans care about?"

Both pipelines must operate at different time scales, use different collection protocols, and feed different decision systems, yet they must integrate at the point where AI proposals are evaluated against reality.

2. Sensor Ground Truth Collection Processes

Ground truth collection establishes the physical and logical reality of network state through deterministic sensor measurements that AI systems cannot hallucinate or approximate.

2.1 Physical Layer Sensor Collection

Optical Transport Network (OTN) Collection Methods

Sensor Type	Collection Protocol	Frequency	Storage Format
Optical Power Meters	SNMP polling of DOM/DDM registers (RFC 2863)	15-second intervals	Time-series DB (InfluxDB/TimescaleDB)
OSNR Sensors	Optical channel monitors via NETCONF/YANG models	30-second intervals	Time-series with spectral metadata
BER Monitors	FEC statistics via gNMI streaming (OpenConfig models)	Real-time streaming (100ms updates)	Apache Kafka topics with time-series archive
PAM4 Modulation Sensors	Transceiver telemetry via OpenConfig terminal-device model	10-second intervals	Signal quality metrics in a graph database
ROADM Monitoring	Per-channel power via ROADM vendor APIs	5-minute intervals	Wavelength topology database

Collection Process Details:

- Data Acquisition Layer:** Dedicated collection agents deployed on network management servers poll transceiver interfaces using SNMP (for legacy equipment) or subscribe to gNMI streams (for modern equipment). Collection agents maintain persistent connections to avoid TCP handshake overhead on high-frequency polling.
- Normalization and Enrichment:** Raw sensor data is normalized to standard units (dBm for power, dB for OSNR) and enriched with topology context from network inventory systems. Each measurement is tagged with fiber span ID, wavelength, direction (eastbound/westbound), and equipment vendor model.
- Quality Validation:** Collection agents implement statistical outlier detection to flag sensor failures. For example, if an optical power reading suddenly changes by >10dB between consecutive samples, the measurement is flagged for manual verification rather than being fed to the Digital Twin.
- Storage and Indexing:** Time-series databases store physical layer telemetry with millisecond-precision timestamps. Indexes are created on (fiber_span_id, timestamp) and (wavelength, timestamp) to enable rapid queries when the Circuit Breaker needs to verify AI proposals.

Critical Implementation Note: Physical layer collection must use out-of-band management networks. If collection traffic traverses the same fibers being monitored, a fiber cut simultaneously disables both the production network and the sensor's visibility into the failure.

2.2 Logical Layer Sensor Collection

5G Core Network Function (NF) Collection Methods

Sensor Type	Collection Protocol	Frequency	Storage Format
NF Status Probes	HTTP health checks to NF management endpoints (3GPP TS 29.510)	5-second intervals	State machine database with event log
CUPS Telemetry	PCFP session monitoring via UPF instrumentation	Real-time event streaming	Kafka with session context preservation
Session Preservation Monitors	UE context tracking via AMF/ SMF event exposure service	Per-event streaming	Graph database with UE lifecycle tracking
gNMI/gRPC Streaming	Microservice mesh telemetry via service proxy sidecars	Sub-second streaming	Distributed tracing backend (Jaeger/Zipkin)

Collection Process Details:

- Service-Based Architecture Integration:** Collection agents subscribe to NF event exposure services defined in 3GPP standards. Each Network Function publishes state changes (registration, de-registration, service updates) to a centralized event bus that sensor collection agents consume.
- State Machine Reconstruction:** The collection system reconstructs the complete state machine for each virtualized network function by correlating health check responses with Kubernetes pod lifecycle events. This enables the detection of scenarios where a container reports 'healthy' to Kubernetes but is actually experiencing internal service degradation.
- Session Continuity Tracking:** For mobile subscriber sessions, collection agents track UE context across handovers, N2 handoffs, and service request procedures. Each session is assigned a globally unique tracking ID that persists across all 5G Core network functions, enabling end-to-end visibility into session preservation failures.
- Cross-Layer Correlation:** Logical layer sensors are correlated with physical layer sensors using timestamps and geographic location tags. For example, if a gNB (base station) reports poor radio conditions and optical sensors simultaneously show degradation on the fiber backhaul to that site, the correlation flags a likely physical infrastructure issue rather than a radio configuration problem.

2.3 Flow Dynamics Collection (Network IMU)

Flow collection captures the 'acceleration' and 'velocity' of traffic patterns, analogous to an Inertial Measurement Unit in robotics.

- **IPFIX/NetFlow Collection:** Routers and switches export flow records every 10-60 seconds containing 5-tuple (source IP, dest IP, source port, dest port, protocol), byte counts, packet counts, and TCP flags. Collectors aggregate these into a flow database indexed by (timestamp, source_prefix, dest_prefix).
- **sFlow Sampling:** For high-speed links (100G+), sFlow samples 1-in-N packets (typically N=4096) to provide statistical traffic visibility without overwhelming collection infrastructure. Sampled packets include full Layer 2-4 headers plus 128 bytes of payload for application classification.
- **BGP/OSPF State Monitoring:** Routing protocol collectors subscribe to BGP Update messages and OSPF LSA floods via BGP Monitoring Protocol (BMP) and passive OSPF adjacencies. This captures routing 'acceleration'—rapid changes in best path selection that indicate network instability.
- **Queue Depth Telemetry:** Switches export egress queue depth metrics every 100ms via streaming telemetry. Sustained queue buildup indicates congestion 'momentum' that precedes packet loss, allowing the Circuit Breaker to detect impending failures before they impact applications.

Critical Architecture Decision: Flow collection infrastructure must be dimensioned for 10x peak traffic burst capacity. During DDoS attacks or flash crowd events, flow export volume can spike by 100x, and collectors that drop flow records during these events lose visibility exactly when it is most needed for Circuit Breaker decisions.

2.4 Digital Twin Integration and Synthesis

Raw sensor data from the physical layer, logical layer, and flow dynamics collection is synthesized into a unified Digital Twin representation.

1. **Graph Database Construction:** The Digital Twin uses a graph model where nodes represent network elements (fibers, transceivers, routers, VNFs) and edges represent relationships (connects_to, carries_traffic, hosts_on). Each node and edge is decorated with time-series properties from sensor collection. (This is what we were trying to construct with the Semantic Governance Program at VZ)
2. **Temporal Consistency Enforcement:** All sensor measurements are tagged with collection timestamps. The Digital Twin maintains a 'current state' view (latest measurement for each sensor) and a historical state index that enables temporal queries like 'What was the OSNR on fiber span X at 14:23:47 UTC yesterday?'

3. **Contradiction Detection:** The Digital Twin continuously runs contradiction detection queries. For example, if a router interface reports 'up' status but optical power sensors show 0 dBm received power, this logical contradiction is flagged as a sensor or equipment failure requiring human investigation.
4. **AI Pre-Flight Check Integration:** When an AI agent proposes a configuration change, it submits a 'proposed graph delta' to the Digital Twin API. The Twin evaluates this delta against the current sensor state and returns a risk score. If the proposal creates paths through degraded links or violates physical constraints (e.g., wavelength contention on ROADMs), the Circuit Breaker vetoes the proposal before execution. (This was part of the service facade we were trying to get funded and built to deliver RT measures)

3. Continuous Value Telemetry Collection Processes

While sensor ground truth establishes what is physically true, value telemetry establishes what is humanly important. (Value measures are derived from Outcome Driven studies, which is a part of customer 360 that we designed into the Semantic Knowledge Layer). This collection pipeline measures drift between operator intent (SLAs, business objectives, regulatory requirements) and AI execution outcomes.

3.1 SLA Monitoring and Collection

Service Level Agreements represent contractual commitments to customers and regulatory obligations. Value telemetry tracks compliance in real-time.

SLA Metric Category	Collection Method	Drift Detection Approach
Availability (Uptime)	Synthetic transaction probes every 1 minute from distributed test points. Each probe attempts service establishment (e.g., TCP 3-way handshake, HTTP GET) and logs success/failure with nanosecond timestamps.	Calculate rolling 5-minute error budget consumption. If AI actions correlate with increased probe failures, the Value-Drift Coefficient increases proportionally.
Latency (Round-Trip Time)	TWAMP (Two-Way Active Measurement Protocol) reflectors at network edges. Test packets stamped with hardware timestamps to achieve microsecond-precision RTT measurement.	Track latency distribution (P50, P95, P99). If AI routing changes cause P99 latency to drift beyond the SLA threshold, Circuit Breaker triggers even if P50 remains compliant.
Packet Loss	Continuous UDP test streams with sequence numbers. Receivers detect gaps in sequence and compute loss rate over 10-second windows.	Correlate loss events with AI configuration changes using temporal alignment. If loss spikes occur within 30 seconds of AI action, causal attribution increases the MTBH score.
Throughput (Goodput)	iPerf3 servers at edge locations accept test connections from distributed clients. Measure achieved throughput vs. theoretical link capacity every 5 minutes.	If AI load-balancing decisions degrade throughput below contracted minimums, calculate revenue impact and escalate to a human trustee for approval override.

Collection Infrastructure Requirements:

- **Probe Diversity:** SLA probes must originate from geographically diverse locations, matching customer population distribution. A probe failure from a single test point might indicate local access issues rather than core network SLA violations.
- **Clock Synchronization:** All SLA collection systems must synchronize to GPS-disciplined PTP (Precision Time Protocol) grandmasters with sub-microsecond accuracy. Without precise timestamps, it is impossible to correlate AI actions with SLA degradation events that occur milliseconds later.
- **Out-of-Band Test Traffic:** SLA test traffic should be tagged with distinct DSCP markings and isolated in separate VLANs to prevent AI traffic engineering algorithms from 'gaming' SLA metrics by prioritizing test traffic over real customer traffic.

3.2 Value-Drift Coefficient Calculation

The Value-Drift Coefficient (Vd) quantifies the divergence between Operator Intent and AI Execution Path in real-time.

Math: $Vd = |Operator_Intent - AI_Execution_Path|$

Where Operator Intent and AI Execution Path are vectorized representations of system state across multiple dimensions: *availability, latency, cost, power consumption, and regulatory compliance*.

Collection Process for Vd Calculation:

1. **Intent Capture:** Operator Intent is formalized in a declarative policy language, which we can publish in an ontology. For example: 'intent: maintain_uptime >= 99.99% for network_slice=emergency_services, priority=highest, cost_sensitivity=none'. These intent statements are versioned in a configuration management database.
2. **Execution Path Observation:** The AI agent's actual configuration decisions are logged with full context: timestamp, proposed change, rationale provided by AI model, and predicted outcome metrics. These logs feed into a distributed tracing system that reconstructs the causal chain of AI decisions.
3. **Semantic Distance Calculation:** A semantic reasoner compares intent policies against actual system behavior using ontological inference. For example, if the intent specifies 'priority=highest' but the AI's routing decision places emergency services traffic on a congested link, the semantic distance calculation detects this misalignment even if the literal configuration syntax is valid.
4. **Real-Time Streaming:** Vd is calculated continuously and published to a Kafka topic within a model-driven service consumed by the Circuit Breaker service. The Circuit Breaker subscribes to Vd events with a tumbling window aggregation (30-second windows) to filter transient spikes from sustained drift.

Example Scenario: An AI agent optimizes for power savings by consolidating traffic onto fewer line cards, reducing active transceivers. However, the Operator Intent specifies 'resilience: n+1_redundancy for all backbone links'. The semantic reasoner detects that consolidation violates n+1 redundancy, calculates $Vd=0.85$ (on a 0-1 scale), and the Circuit Breaker halts the consolidation before it executes.

3.3 MTBH (Mean Time Between Hallucinations) Collection

MTBH measures how frequently an AI agent makes confident predictions that contradict sensor ground truth, providing a reliability metric for AI decision quality.

Hallucination Detection Process:

1. **Prediction Logging:** Every AI prediction is logged with a timestamp, confidence score (0-1), and predicted outcome. For example: 'prediction: path_A_stable=true, confidence=0.94, timestamp=2025-02-07T14:23:00Z'.

2. **Ground Truth Verification:** Within 60 seconds of the prediction, the system queries sensor ground truth from the Digital Twin. For the path stability example, it retrieves actual BER, OSNR, and packet loss measurements for the predicted path.
3. **Contradiction Scoring:** If the prediction contradicts sensor measurements (e.g., AI predicted 'stable' but sensors show BER > threshold), the system logs a hallucination event. Contradiction severity is weighted by AI confidence: high-confidence wrong predictions are worse than low-confidence errors.
4. **MTBH Calculation:** MTBH is computed as a sliding window metric: total elapsed time divided by count of hallucination events in the window. For example, MTBH=25 hours means the AI makes one confidently wrong prediction every 25 hours on average.
5. **Categorization by Decision Type:** MTBH is tracked separately for different AI decision categories: routing changes, capacity scaling, failover triggers, etc. This enables the Circuit Breaker to apply different trust thresholds based on the type of decision the AI is proposing.

Storage and Indexing: Hallucination events are stored in a time-series database with rich metadata aligned/mapped to an ontology framework for consistency of meaning: AI model version, input features used, predicted vs. actual outcomes, and environmental context at prediction time. This historical database enables root cause analysis of AI failures and model retraining prioritization.

3.4 Cognitive Load Monitoring for Human Trustees

As AI systems become more sophisticated, they may propose changes that are technically sound but difficult for humans to verify. Cognitive load monitoring protects against trustee overwhelm.

- **Decision Complexity Metrics:** Each AI proposal is scored for complexity: number of affected network elements, cross-layer dependencies, number of simultaneous changes, and historical precedent. A proposal affecting 500 routers across 3 metro areas with no historical precedent receives a high complexity score.
- **Queue Depth Tracking:** The system tracks how many AI proposals are currently awaiting trustee review, their arrival rate, and trustee response times. If the queue depth exceeds trustee capacity (e.g., >10 pending reviews for a single trustee), new low-priority proposals are automatically deferred.
- **Fatigue Detection:** Trustee interaction patterns are analyzed for signs of decision fatigue: decreasing time spent reviewing proposals, increasing approval rate without complexity decrease, or longer response latencies. When fatigue indicators exceed thresholds, the system reduces the AI proposal rate.
- **Adaptive Circuit Breaker Sensitivity:** When cognitive load is high, the Circuit Breaker automatically becomes more conservative, lowering the Vd threshold for autonomous

approvals and escalating more decisions to human review. This prevents trustees from rubber-stamping approvals under time pressure.

4. Collection Integration and System Architecture

Sensor ground truth and value telemetry collection pipelines must integrate at the Circuit Breaker decision point, where AI proposals are evaluated.

4.1 Data Pipeline Architecture

The collection architecture follows a lambda architecture pattern with separate batch and streaming processing paths.

- **Streaming Path (Hot):** Real-time sensor data flows through Apache Kafka topics, processed by Apache Flink stream processors that calculate windowed aggregates (5-minute OSNR averages, 1-minute flow throughput, etc.). Results feed directly to the Circuit Breaker API for pre-flight checks.
- **Batch Path (Cold):** Historical sensor data is archived in Parquet files in object storage (S3/Azure Blob). Nightly Spark jobs compute statistical baselines: typical OSNR ranges per fiber span, normal traffic variance by time of day, SLA compliance trends. These baselines inform anomaly detection thresholds.
- **Serving Layer:** The Digital Twin graph database serves as the unified query interface for both hot and cold data. Time-series databases (InfluxDB) cache recent sensor readings for microsecond-latency queries, while graph queries span historical context from the batch layer.

4.2 The Semantic Handshake Protocol

Before any AI-proposed change executes, it must pass a Semantic Handshake that verifies alignment between probabilistic intent and deterministic reality.

Handshake Sequence:

1. **AI Proposal Submission:** The AI agent submits a structured proposal to the Circuit Breaker API: {action: 'route_traffic', source: 'AMF-Chicago', dest: 'SMF-Detroit', capacity: '1Tbps', rationale: 'lower_latency_path', confidence: 0.89}.
2. **Ground Truth Query:** Circuit Breaker queries Digital Twin for current state: 'SELECT osnr, ber, cpu_utilization FROM network_path WHERE source=AMF-Chicago AND dest=SMF-Detroit'. Returns sensor measurements timestamped within the last 30 seconds.
3. **Delta Calculation:** Circuit Breaker computes the delta between AI assumptions and sensor reality. For example, AI assumes SMF-Detroit has <80% CPU utilization, but sensors show 87% actual utilization. $\text{Logical_Delta} = |\text{intended_headroom} - \text{actual_headroom}| = 13\%$.

4. **Value-Drift Check:** Simultaneously, Circuit Breaker retrieves the current Vd score for the affected service slice. If Vd is already elevated due to prior AI decisions, this new proposal is subject to stricter scrutiny.
5. **Threshold Evaluation:** If (Logical_Delta > Threshold_L) OR (Physical_Delta > Threshold_P) OR (Vd > Threshold_V), then Circuit Breaker initiates Deterministic Veto and the proposal is rejected. Otherwise, the proposal is auto-approved for execution.
6. **Human Escalation:** Rejected proposals are logged with full context and optionally escalated to human trustees if AI confidence was high (>0.85) despite veto. This creates a feedback loop where humans review edge cases and adjust thresholds.

Handshake Latency Target: *The entire Semantic Handshake must complete in <100ms to avoid introducing unacceptable latency in network automation workflows. This requires careful optimization of Digital Twin query performance and Circuit Breaker threshold evaluation logic.*

4.3 Collection System Scalability and Reliability

Enterprise-scale telecommunications networks generate petabytes of sensor and telemetry data annually. The collection infrastructure must scale horizontally while maintaining reliability.

- **Distributed Collection Agents:** Sensor collection is distributed across geographic regions with local aggregation before centralized storage. Each metro region runs a collection cluster that pre-processes sensor data, computes local aggregates, and forwards only summaries to the central data lake.
- **Message Queue Durability:** Kafka topics use replication factor=3 with min.insync.replicas=2 to ensure no sensor data is lost even during broker failures. Critical telemetry streams (SLA metrics, Vd calculations) use synchronous replication with acknowledgment.
- **Database Partitioning:** Time-series databases partition data by (network_region, timestamp) to enable parallel query execution. Graph databases shard by network layer (physical, logical, service) to distribute query load across clusters.
- **Data Retention Policies:** Raw sensor data: 90 days. Aggregated statistics: 2 years. Hallucination events and Circuit Breaker decisions: 7 years (regulatory compliance). Retention policies are enforced through automated TTL expiration in databases.

5. Implementation Roadmap and Maturity Assessment

Organizations should adopt collection processes incrementally, validating each stage before proceeding to the next level of maturity.

Maturity Levels:

- **Level 1 - Basic Sensor Deployment:** Install optical power monitors and basic flow collectors. Establish time-series databases and validate data quality. Success metric: 95% sensor uptime, <1% data loss rate.
- **Level 2 - Digital Twin Integration:** Build a graph database representation of network topology. Integrate sensor data into the Digital Twin model. Success metric: Query latency <50ms for 95th percentile, complete topology coverage.
- **Level 3 - SLA and Value Telemetry:** Deploy synthetic monitoring for SLA tracking. Implement the Vd calculation pipeline. Success metric: SLA probe coverage across all customer-facing services, Vd updates every 30 seconds.
- **Level 4 - Circuit Breaker Integration:** Implement the Semantic Handshake protocol. Connect AI decision APIs to Circuit Breaker evaluation. Success metric: 100% of AI proposals verified against ground truth before execution, handshake latency <100ms.
- **Level 5 - Adaptive Trust Optimization:** Deploy MTBH tracking and cognitive load monitoring. Enable dynamic threshold adjustment based on AI reliability history. Success metric: Zero critical service degradations caused by AI decisions, trustee approval rate <5% of total AI actions.

Summary

Trustworthiness Engineering in telecommunications depends fundamentally on robust collection processes for sensor ground truth and continuous value telemetry. Without high-fidelity, real-time visibility into both physical reality and business value alignment, AI agents operate blindly—potentially making confident but catastrophic decisions.